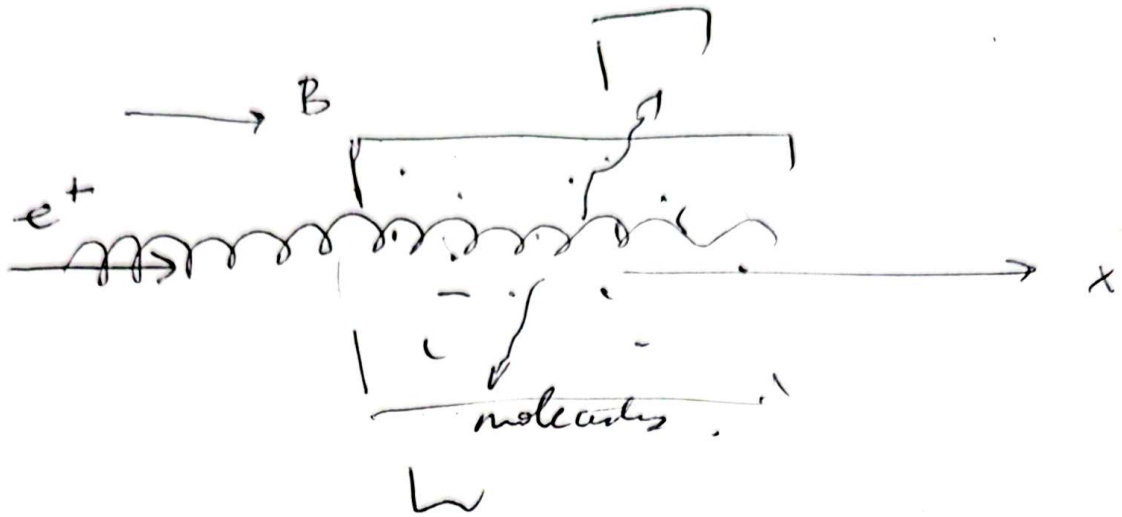
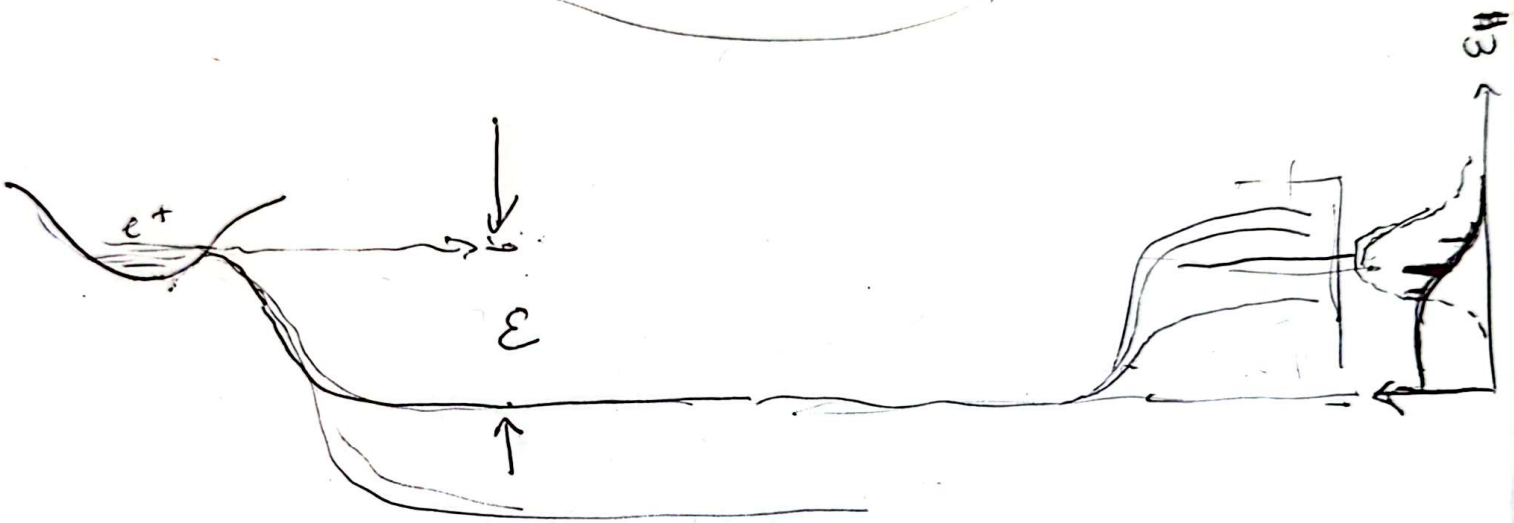


$$\underline{\underline{\mathcal{E}}} = \mathcal{E}_{\parallel} + \mathcal{E}_{\perp} = \frac{m v_z^2}{2} + \frac{m(v_x^2+v_y^2)}{2}$$



$\mathcal{E}_{\perp}$ has a Maxwellian ~~con~~ velocity distribution

$$e^{-\frac{m(v_x^2+v_y^2)}{2kT}} \, dv_x \, dv_y$$

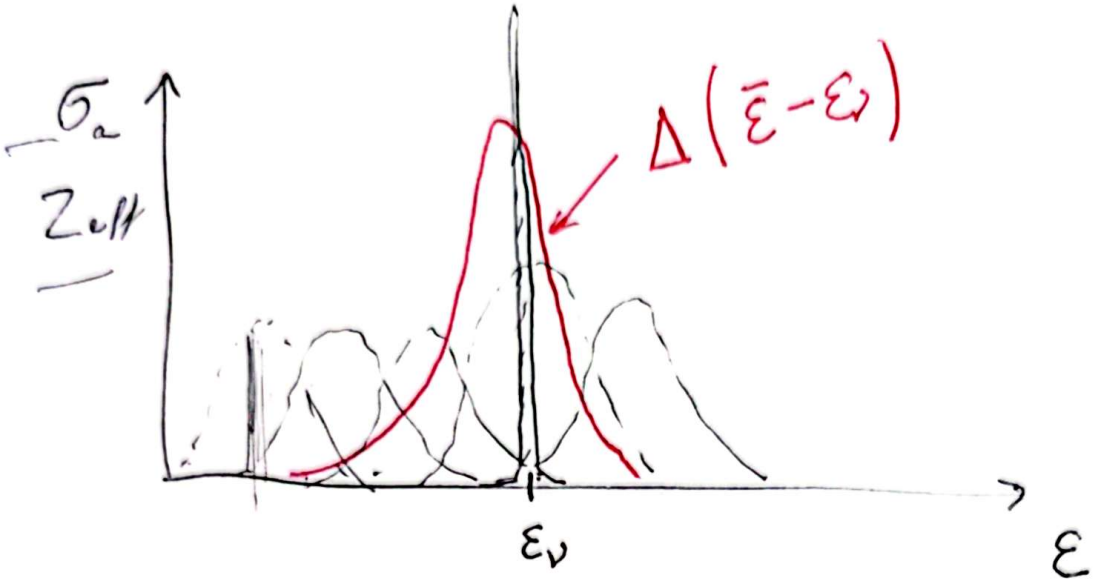
$\mathcal{E}_{\parallel}$ has an approximate Gaussian distribution

$$e^{-\frac{(\mathcal{E}_{\parallel}-\bar{\mathcal{E}})^2}{2\sigma^2}}$$

$$f_{\bar{\mathcal{E}}}(\mathcal{E}_{\parallel}, \mathcal{E}_{\perp}) = A \, e^{-\frac{\mathcal{E}_{\perp}}{kT}} \, e^{-\frac{(\mathcal{E}_{\parallel}-\bar{\mathcal{E}})^2}{2\sigma^2}}$$

pos. en. prob. dist. f'n.

$\bar{\mathcal{E}}$: mean parallel energy.



ε_ν the energy of the resonance

$$= \int \delta(\varepsilon - \varepsilon_\nu)\, f_{\bar{\varepsilon}}(\varepsilon_{\parallel}, \varepsilon_{\perp})\, d\varepsilon_{\parallel}\, d\varepsilon_{\perp}$$

$$\Delta(\bar{\varepsilon} - \varepsilon_\nu)$$

Breit-Wigner formula

$$\sigma = \frac{\pi}{k^2} \frac{\Gamma_\nu^{(a)} \Gamma_\nu^{(e)}}{(\varepsilon - \varepsilon_\nu)^2 + \frac{\Gamma_\nu^2}{4}}$$

out in

$e^{i\bar{k}\cdot\bar{r}}$

$$e^+ + M \underset{\leftarrow}{\overset{\Gamma_\nu^{(e)}}{\rightarrow}} e^+ M(\nu) \xrightarrow{\Gamma_\nu^{(a)}} M^+ + 2\gamma$$

$$\Gamma_\nu = \Gamma_\nu^{(e)} + \Gamma_\nu^{(a)}$$

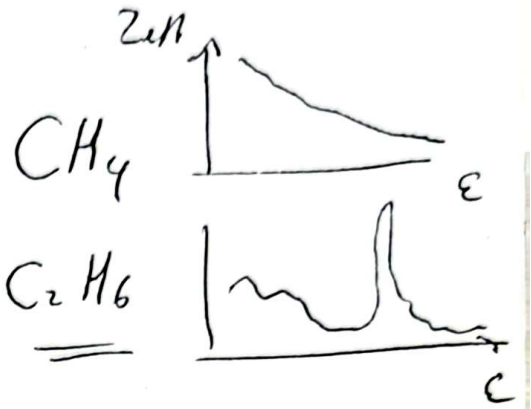
$$\boxed{\mathcal{E}_\nu = \hbar\omega_\nu - \mathcal{E}_b}$$

No. of modes :

$3N - 6$

$\uparrow$

no. of. atoms.



CH_3F

Cl

Br

I

$$\mathcal{E}_\nu = \sum_i n_i \hbar\omega_i - \mathcal{E}_b$$

Web of Science

PRA

$\int |\psi_0(\vec{r})|^2 d\vec{r} = 1$

$r^2 \sin\theta \, dr \, d\theta \, d\phi$

2017 :

theo + exp.

John Stanton

2 - quantum

+ 3 - quantum

$-\frac{\varepsilon_b}{2} = -\mathcal{E}_b$

$x \ll 1$

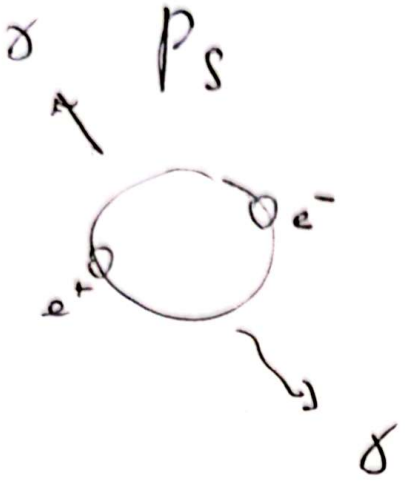
(in atomic unit)

$R_a \sim 1$

$$\psi_0(\vec{r}) \sim A \frac{e^{-x r}}{r}$$

$$-\frac{1}{2}\nabla^2\psi + U(\vec{r})\psi(\vec{r}) = E\,\psi(\vec{r})$$

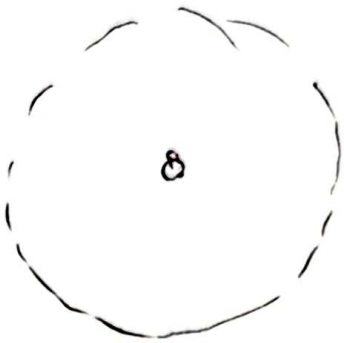
$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$



$S = 0$

$\tau = 0.125\ ns \sim 10^{-10}\ s.$

$\tau_{at} = \frac{R_a}{v_a} \sim 10^{-17}\ s.$



10^7 atom year

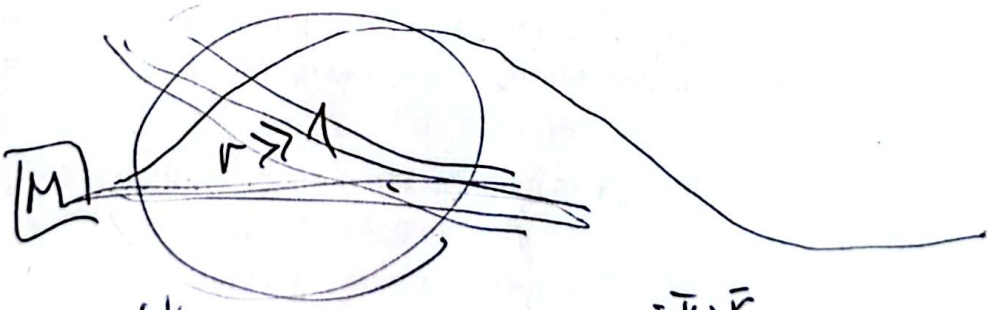
$\Delta E\, \tau \sim \hbar = 1$

$\Delta E\, \tau \sim 1$

$\Gamma \tau \sim 1$

dipole moment

$$V(\vec{r}) = \frac{\vec{d} \cdot \vec{r}}{r^3}$$



ψ_0 $\qquad \psi_{\vec{k}} = e^{i\vec{k}\cdot\vec{r}}$

$k \ll 1$